

Physico-mechanical properties of soil

- Introduction to soil mechanics and geotechnical engineering
- Origin of soils
- Basic relationships
- Index properties including consistency and gradation
- Clay mineralogy
- Classification of engineering soils
- Engineering properties of soils
- compressibility, consolidation, compaction and strength.

Geotechnical Engineering ~ Soil Mechanics

- Geotechnical engineering is concerned with the application of civil engineering technology to some aspect of the earth, usually the natural materials found on or near the earth's surface.

Soil, in an engineering sense; is the relatively loose agglomerate of mineral and organic materials and sediments found above the bedrock.

The dividing line between soil and rock is arbitrary, and many natural materials encountered in engineering practice cannot be easily classified. They may be either a "very soft rock", or a "very hard soil"



- In geology, for example, rock means all the materials found in the earth's crust, including what most of us would call soil.
- Soils to a geologist are just decomposed and disintegrated rocks found in the very thin upper part of the crust and usually capable of supporting plant life.
- Similarly, pedology (soil science) and agronomy are concerned with only the very uppermost layers of soil - that is, those materials important to agriculture and forestry.

Geotechnical engineering has several different aspects or emphases.

- Soil mechanics is concerned with the engineering mechanics and properties of soil,
 - whereas rock mechanics is concerned with the engineering mechanics and properties of rock---: usually, but not limited to, the bedrock.
 - Soil mechanics applies to soils the basic principles of mechanics including kinematics, dynamics, fluid mechanics, and the mechanics of materials.
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- soil masses behave differently from rock masses, in practice there is not much overlap between the two disciplines

Inconveniently, the world does not consist only of soft or loose soils and hard rock, but rather, most geo-materials fall somewhere between those extremes. In a professional practice one will have to learn to deal with a wide range of material properties and behaviors.

Foundation engineering applies

- engineering geology
- soil mechanics
- rock mechanics, and
- structural engineering

to the design and construction of foundations for civil engineering and other structures.

The foundation engineer must be able to predict the performance or response of the foundation soil or rock to the loads the structure imposes.



Examples include

- foundations for industrial
- commercial, and residential buildings
- Bridges
- towers,
- retaining walls
- foundations for oil and other kinds of tanks and offshore structures
- Ships must have a drydock during construction or repairs, and the drydock must have a foundation.
- During construction and launch, rockets and appurtenant structures must be safely supported.



- Other related geotechnical engineering problems the foundation engineer faces are
 - stability of natural and excavated slopes
 - stability of permanent and temporary earth-retaining structures,
 - control of water movement and water pressures
 - maintenance and rehabilitation of old buildings.

Not only must the foundation safely support static structural and construction loads, but it must also adequately resist dynamic loads due to wind, blasting, earthquakes, and the like.

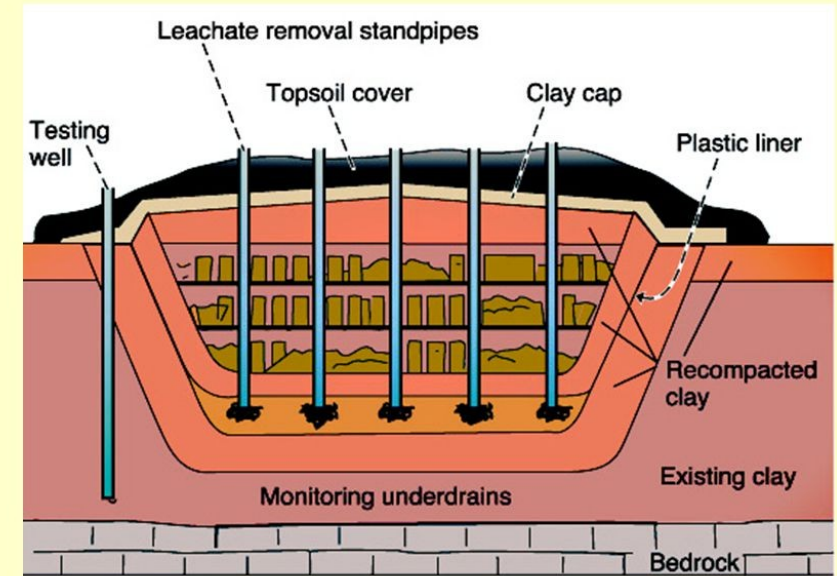
GEOENVIRONMENTAL ENGINEERING OR ENVIRONMENTAL GEOTECHNICS

Challenges faced

- problems of polluted groundwater
- proper disposal and containment of municipal and industrial wastes
- design and construction of nuclear waste repositories
- remediation of hazardous waste repositories (aka dumps) and other contaminated sites.

Although all these problems have a major geotechnical engineering component, they are interdisciplinary in nature, and their solutions require that geotechnical engineers work together with environmental and chemical engineers, environmental and public health specialists, geohydrologists, and regulatory agency personnel.

Secure Landfills

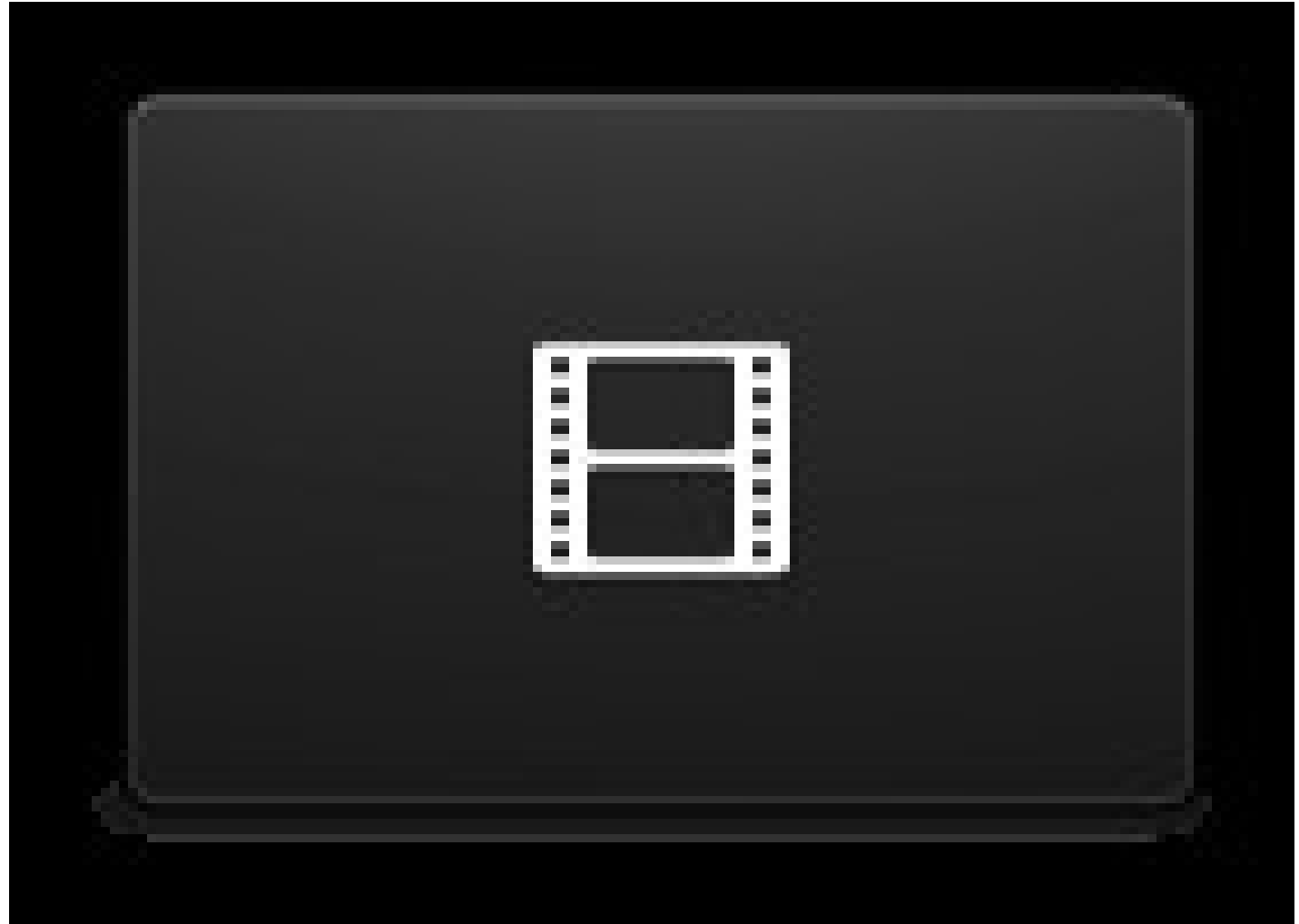


HISTORICAL DEVELOPMENT OF GEOTECHNICAL ENGINEERING

- The ancient Egyptians, Babylonians, Chinese, and Indians knew about constructing dikes and levees / dams out of the soils found in river flood plains
- European architects and builders during the Middle Ages learned about the problems of settlements.
- The tower began to lean during construction in the 12th century, due to soft ground which could not properly support the structure's weight, and it worsened through the completion of construction in the 14th century. By 1990, the tilt had reached 5.5 degrees. The structure was stabilized by remedial work between 1993 and 2001, which reduced the tilt to 3.97 degrees.
- Vikings in Scandinavia used timber piles to support houses and wharf structures on their soft clay.



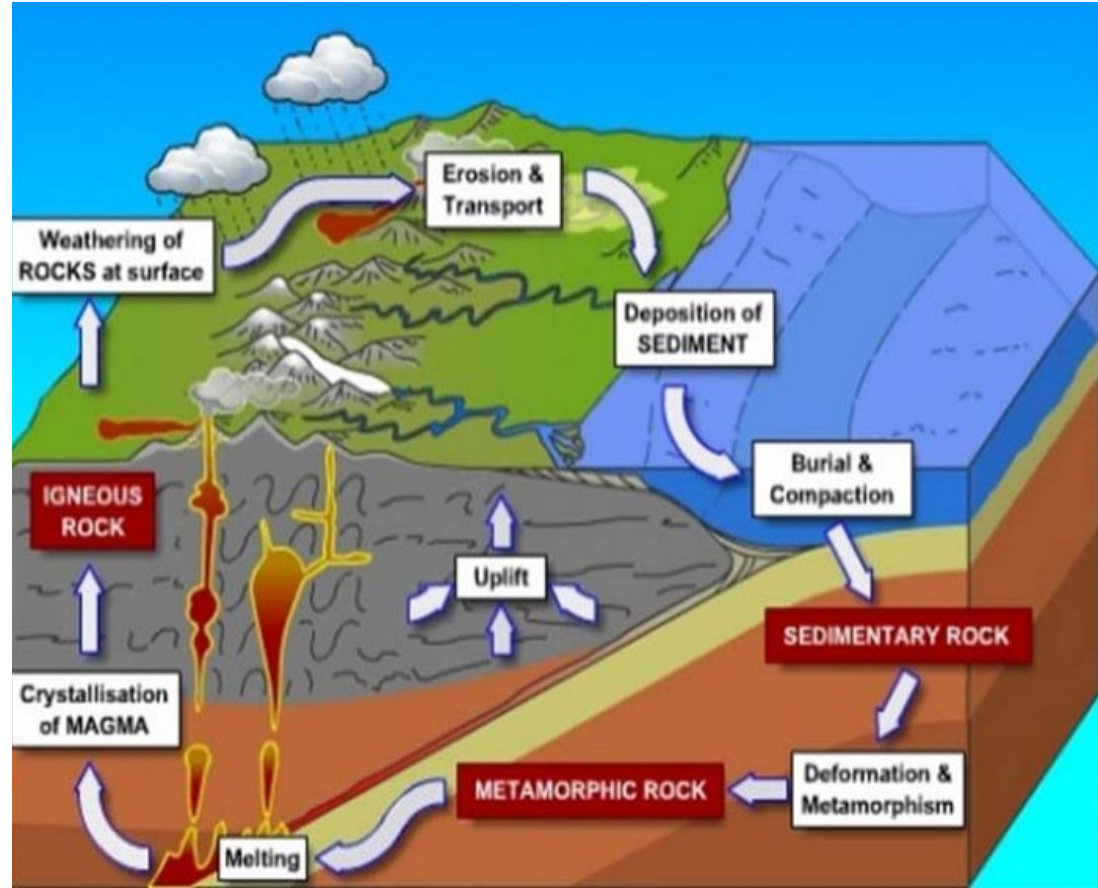
Driving
piles in
soft clay



- The "design" of foundations and other constructions involving: soil and rock; was by rule of thumb, and very little theory as such was developed until the mid~1700s.
- Coulomb is the most famous 'engineering name of that era. He investigated the problems of earth pressures against retaining walls, and some of his calculation procedures are still in use today. The most common theory for the shear strength of soils is named after him (Coulomb, 1776)
- Collin (1846) was the first engineer to systematically examine failures in clay slopes as well as the measurement of the' shear strength of clays.
- Darcy (1856) established his law for the flow of water through sands.
- Rankine (1857) developed a method for estimating the earth pressure against retaining walls.
- Atterberg (1911) defined consistency limits for clays that are still in use today.
- Geotechnical Commission of the Swedish State Railways (1914-1922): developed many important concepts and apparatuses in geotechnical engineering

- with these early developments in Sweden'; the true father of soil mechanics is an Austrian, Prof. Karl Terzaghi. He published the first modern textbook on soil mechanics in 1925, and in fact the name "soil· mechanics" is a translation of the German word Erdbaumechanik; which was part of the title of that book (Terzaghi, 1925a).
- Another important figure is Prof. Arthur Casagrande,' who was at Harvard University from 1932 until 1969.

- In general soils are formed by weathering of rocks
- The physical properties of a soil are dictated by primarily by the minerals that constitute the soil particles and, hence, the rock from which it is derived.
- Many of the physical properties of the soil are dictated by the size, shape, and chemical composition of the grains.
- Soil is formed by geological cycle which consist Erosion, Transportation, Deposition and Upheaval.



Typical Products of Weathering

Rock Type		Soil Type
Granite	→	Silty Sands
Basalt	→	Clayey Soil
Shale / Slate	→	Clay / Silt
Sandstone	→	Sand
Limestone	→	Clay / Silt

The transported soil may be classified into several groups, depending on their mode of transportation and deposition.

Transporting Agent

Soil Type

Gravity



Colluvial

Water



Alluvial

Streams

Fluvial

Rainfall

Pluvial

Ice



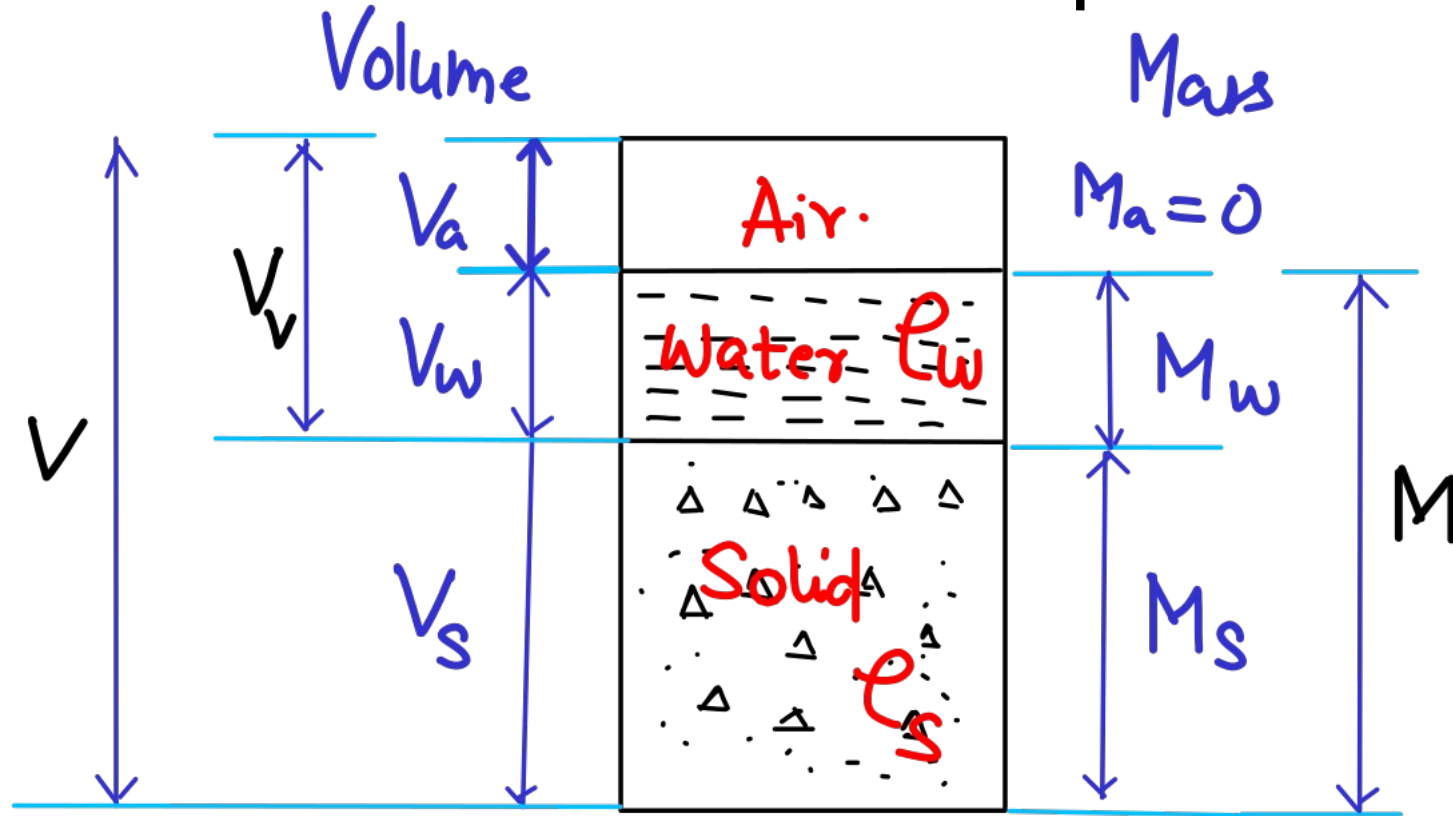
Glacial

Wind



Aeolian

Basic relationships



Already Discussed in Chapter -5

$$(1) \text{ Water Content } (w) = \frac{M_w}{M_s} \times 100 \quad (\%)$$

$$(2) \text{ Degree of Saturation } (S_r) = \frac{V_w}{V_v} \times 100 \quad (\%)$$

$$(3) \text{ Porosity } (n) = \frac{V_v}{V} \times 100 \quad (\%)$$

$$(4) \text{ Void ratio } (e) = \frac{V_v}{V_s} \quad (\text{Ratio})$$

$$(5) \text{ Density / Bulk density / mass density } (\rho) = \frac{M}{V} = \frac{M_s + M_w}{V} \quad (\text{kg/m}^3)$$

$$(6) \text{ Relative density } (d) \quad (\text{sp. gr.}) = \frac{\rho}{\rho_w} \quad (\text{No unit})$$

$$(7) \text{ Dry Density } (\rho_d) = \frac{M_s}{V} \quad (\text{kg/m}^3)$$

$$(8) \text{ Saturated Density } (\rho_s) = \frac{M_s + V_v \rho_w}{V} \quad (\text{kg/m}^3)$$

$$V_v = V_w \quad (\text{when \% Saturated \& } V_a = 0)$$

$$(9) \text{ Saturated Relative Density } d_{\text{sat}} = \frac{\rho_s}{\rho_w} \quad (\text{No unit})$$

(Saturated sp.gr)

$$(10) \text{ Grain density } (\rho_s) = \frac{M_s}{V_s} \quad (\text{kg/m}^3)$$

(density of Solids)

$$(11) \text{ Grain Relative density } (d_s) \quad = \quad \frac{\rho_s}{\rho_w} \quad (\text{No unit}).$$

(grain sp. gr.)

$$(12) \text{ Unit weight } (\gamma) \quad = \quad \rho \cdot g \quad (\text{N/m}^3).$$

Problem:

Given :

$$\rho = 1.76 \text{ Mg/m}^3 \text{ (total density or bulk density)}$$

$$w = 10 \% \text{ (~~total density or bulk density~~)}$$

$$\rho_s = 2.7 \text{ Mg/m}^3 \text{ (grain density or density of solid)}$$

Determine:

$$\rho_d = \text{Dry density}$$

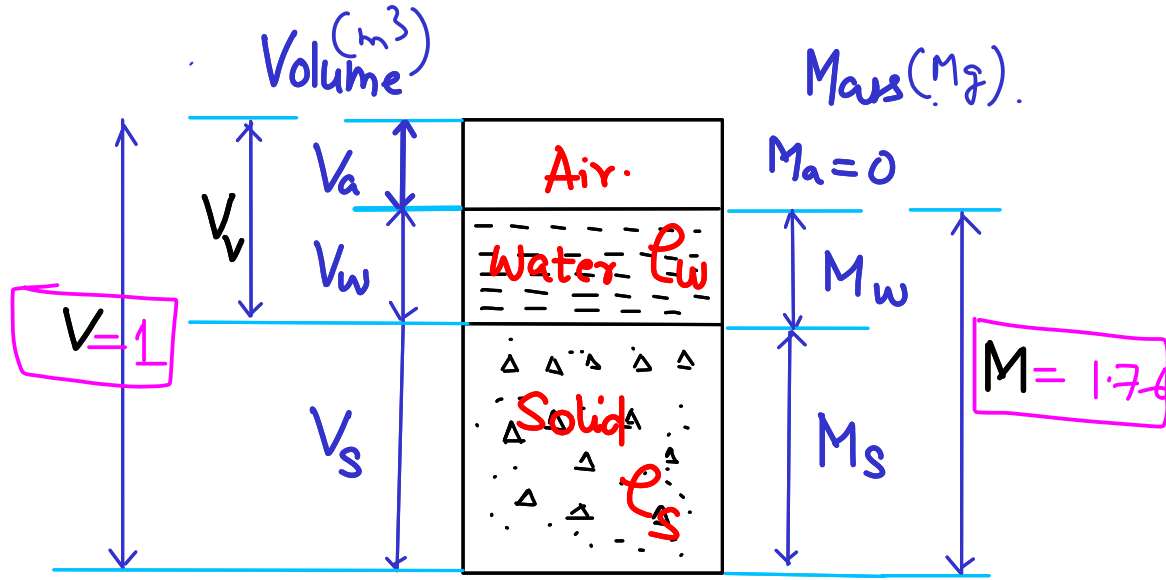
$$e = \text{Void ratio}$$

$$n = \text{porosity}$$

$$S_r = \text{Degree of saturation}$$

$$\rho_{sat} = \text{Saturated density}$$

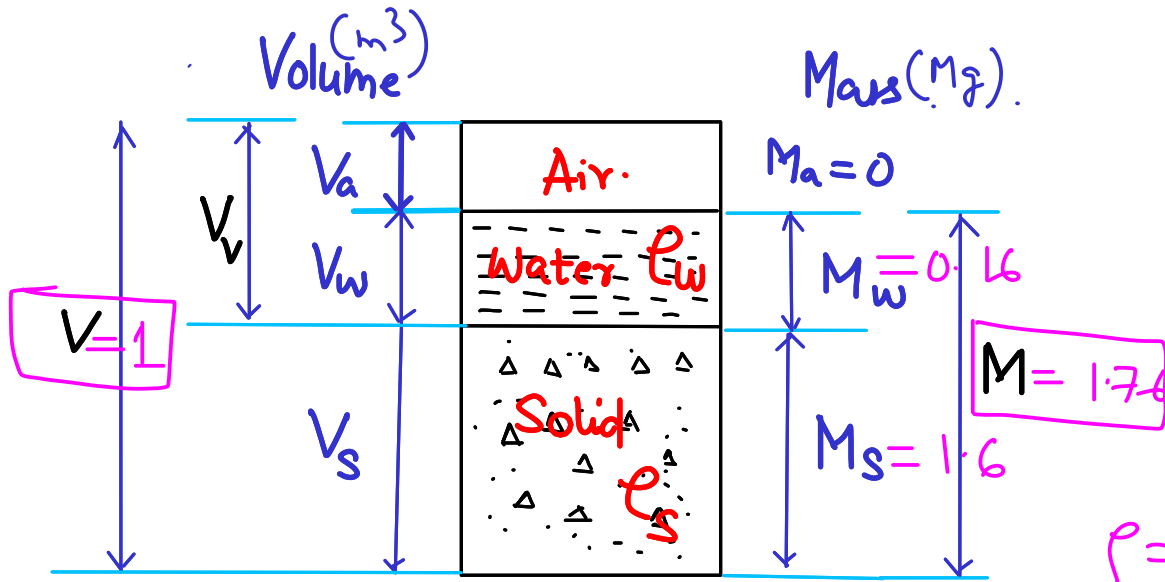
Solution.



Given

$$\rho = 1.76 \text{ Mg/m}^3$$

$$\begin{aligned} \text{Let } V_d &= 1 \text{ m}^3 \\ \Rightarrow M &= 1.76 \end{aligned}$$



Step-2

$$\omega = 10\%$$

$$0.1 = \frac{M_w}{M_s}$$

$$M_w = 0.1 M_s \quad (1)$$

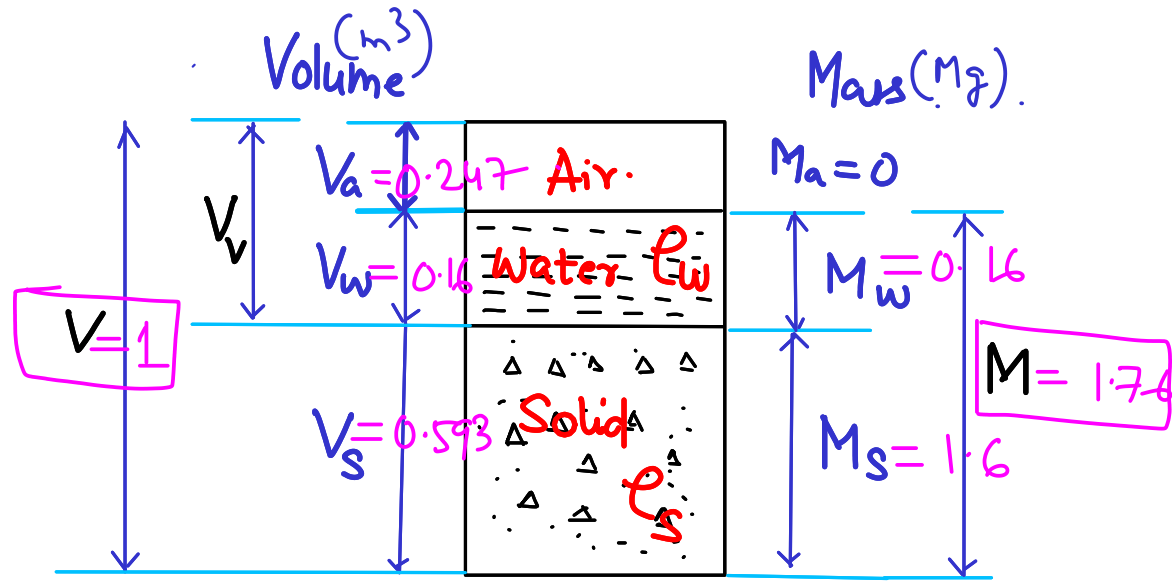
$$\rho = 1.76 \text{ Mg/m}^3 = \frac{M}{V} = \frac{(M_w + M_s)}{V} \text{ Mg}$$

$$1.76 = \frac{0.1 M_s + M_s}{1}$$

$$\Rightarrow M_s = \frac{1.76}{1.1} = 1.6 \text{ Mg}$$

$$M_w = 0.16 \text{ Mg}$$

ply the

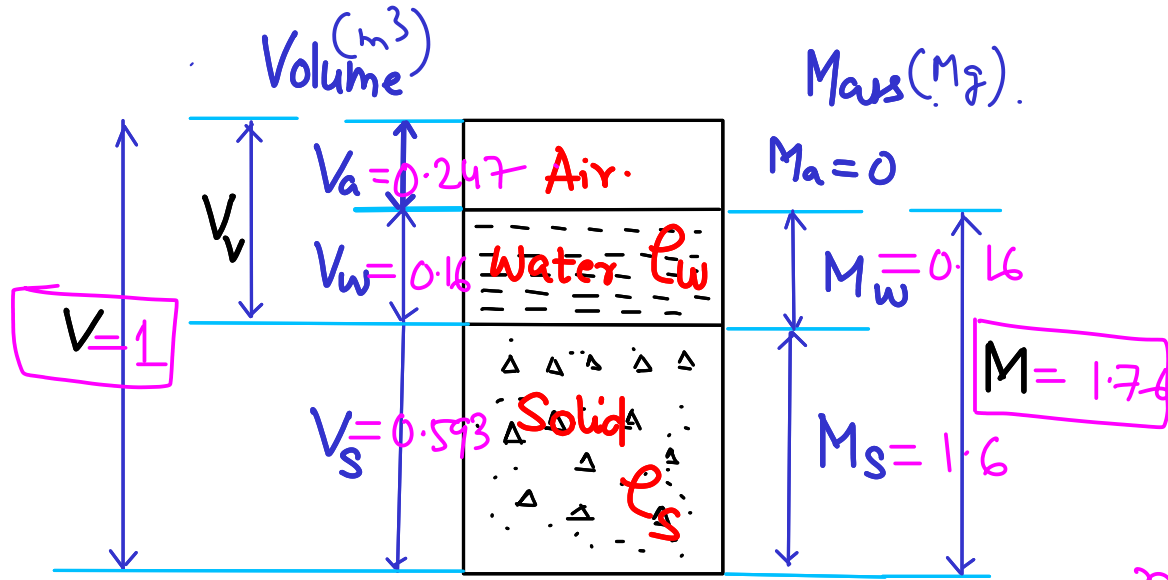


$$\rho_w = \frac{M_w}{V_w}$$

$$V_w = \frac{M_w}{\rho_w} = \frac{0.16 Mg}{1 Mg/m^3} = 0.16 m^3 \quad \leftarrow \rho_{H_2O}$$

$$\rho_s = \frac{M_s}{V_s} \Rightarrow V_s = \frac{M_s}{\rho_s} = \frac{1.6 Mg}{2.7 Mg/m^3} = 0.593 m^3$$

$$V_a = 1 - 0.16 - 0.593 = 0.247 \quad \leftarrow \rho_{H_2}$$



$$\rho_a = \frac{M_s}{V} = \frac{1.6}{1} \text{ Mg/m}^3$$

$$e = \frac{V_v}{V_s} = \frac{0.16 + 0.247}{0.593} = 0.686$$

$$n = \frac{V_v}{V} = \frac{0.16 + 0.247}{1} \times 100 = \underline{\underline{40.7\%}}$$

$$S = \frac{V_w}{V_v} = \frac{V_{wi}}{V_c + V_w} \times 100 = \frac{0.16}{0.247 + 0.16} \times 100$$

$$= 0.3931 \times 100$$

$$= \underline{39.31 \%}$$

$$\rho_{\text{sat}} = \frac{M_w + M_s}{V} = \frac{\rho_w \times V_w + M_w + M_s}{V}$$

$$= \frac{(0.247 + 0.16 + 1.6) \text{ Mg}}{1 \text{ m}^3}$$

$$= \underline{2.01 \text{ Mg/m}^3}$$

